

1. A study is commissioned to examine a possible link between snoring and sleeping position. The results of examining 60 volunteers in a sleep clinic are below:

	Back Sleeper (B)	Side Sleeper (S)	Tummy Sleeper (T)	Total
Does Not Snore (X)	10	22	11	43
Snore (Y)	8	7	2	17
Total	18	29	13	60

Suppose that one case is selected at random from this study.

- (a) (1 point) Find the probability that a participant in this study snores.
 - (b) (1 point) Find the probability that a participant snores, given that they are a back sleeper.
 - (c) (2 points) Find $P(Y \text{ or } T)$.
 - (d) (2 points) Find $P(X^C \text{ and } B)$.
 - (e) (2 points) Are the events “snore” and “back sleeper” independent? Justify your answer mathematically.
2. People are asked to roll two non-standard dice: one is a six-sided die with 2 red faces and 4 blue faces, the other is a ten-sided die with 3 yellow faces and 7 purple faces.
- (a) (1 point) Suppose a person rolls the pair of dice once. What is the probability of rolling a red?
 - (b) (1 point) Suppose a person rolls the pair of dice once. What is the probability of someone rolling a red and a yellow?
 - (c) (4 points) Suppose a person rolls the pair of dice twice. What is the probability of rolling red and yellow twice or blue and purple twice?
 - (d) (3 points) Suppose a person rolls the pair of dice eight times. What is the probability that they will roll a red exactly two times?
 - (e) (3 points) Suppose a person rolls the pair of dice eight times. What is the probability that they will roll a red two times or more?
3. (2 points) An inn has 12 empty rooms and 8 separate guests arriving today. In how many ways can room assignments be made today?
4. Your little cousins make odd pizzas by drawing ingredient names from a hat without replacement. There are 11 possible ingredients to be drawn: pepperoni, mushrooms, pineapple, oysters, chocolate chips, apple slices, artichoke, spinach, olives, sausage, and hot peppers.
- (a) (2 points) If they choose six ingredients at random for their next pizza, how many possible pizzas can be created?
 - (b) (2 points) What is the probability that the six-ingredient pizza just mentioned will contain both pepperoni and chocolate chips?
5. Dan works in a busy restaurant. When he has a good shift, he is 70% likely to stay and have a meal with his colleagues once he's off the clock. If he has a bad shift, he is 40% likely to stay for a meal. He likes his job and claims that 80% of his shifts are good.

- (a) (4 points) Find the probability that Dan does NOT stay after his shift for a meal.
- (b) (4 points) Tonight, Dan is staying for a meal. Calculate the probability that he had a good shift.
6. We study the packing habits of 500 randomly selected passengers on commercial flights between Canada and the USA in 2019. The number of bags that they travelled with (including personal items) were counted. The results are as follows:

Number of bags (x)	0	1	2	3	4
Probability ($P(x)$)	0.012	0.266	0.632	?	0.038

- (a) (1 point) Find the missing probability.
- (b) (1 point) What is the probability of someone travelling with at least two bags?
- (c) (2 points) Calculate the expected number of bags per air traveller.
7. A company analyzes employment records and finds that the amount of time an employee will remain at the company is normally distributed with a mean length of 8.1 years and a standard deviation of 2.9 years.
- (a) (4 points) Calculate the probability that an employee will remain at the company for 8.5 to 10 years.
- (b) (4 points) If 9% of employees have been at the company longer than Antoine, how long has Antoine been with the company?
8. On an investment platform, an analysis of last year's results revealed that each dollar invested gave an earning that is normally distributed with an average of \$0.094 and a population standard deviation of \$0.046.
- (a) (3 points) Calculate the probability that one dollar invested yielded a return of at least \$0.08.
- (b) (1 point) We wish to find the probability that tracking the value of \$100 randomly invested dollars on this platform yielded a return of at least \$8 last year, that is, \$0.08 per dollar invested. Briefly explain why we can apply the CLT to this situation.
- (c) (3 points) Use the CLT to determine the probability that the value of \$100 randomly invested dollars on this platform yielded a return of at least \$8 last year.
9. In 2026, 22% of employed Canadians work in the public sector. A room is filled with a random sample of 40 employed Canadians. Let r represent the number of individuals in that room who work in the public sector.
- (a) (1 point) Calculate the expected value of r .
- (b) (1 point) Calculate the standard deviation of r .
- (c) (4 points) Use a normal approximation to estimate the probability that at least ten individuals in the room work in the public sector.
10. We wish to establish what percentage of Canadians suffer from Seasonal Affective Disorder (SAD). Rather than simply asking if people suffer from SAD, a questionnaire is to be administered to a random sample of Canadians. The results of the questionnaire will allow participants to be categorized as "suffering from SAD" and "not suffering from SAD."

- (a) (4 points) How many Canadians would have to complete the questionnaire if we want our estimate to be within 4% of the true proportion at a 94% confidence level?
- (b) (3 points) Suppose that with only 120 responses, we find that 8 participants qualified as SAD-sufferers. Construct a 99% confidence interval for the true population proportion of SAD-sufferers.
- (c) (1 point) Circle the description below that best interprets the meaning of the confidence interval found in part (b).
- A. The confidence interval is 99% effective at identifying the true population proportion.
 - B. There is a 99% probability that the true population mean is in the confidence interval.
 - C. If we repeat the process of finding a similar confidence interval with different random samples, the true population proportion should lie in the resulting confidence interval 99% of the time.
- 11.** (4 points) An international partnership between environmental scientists hopes to determine the net proliferation rates of arctic fox pups. Tracking a random sample of 41 arctic fox litters, it is found that the average number of pups per litter who survive until the following year is 2.6 with a sample standard deviation of 0.8. Construct a 95% confidence interval for the true mean number of surviving pups per litter.
- 12.** (5 points) Social psychologists wish to test whether or not there is any truth to the claim that adrenaline alters the speed at which bonding occurs in romantic relationships. The metric used to confirm quick bonding in this study was a rapid decision to go on another date.
- A simple random sample of couples on first dates with exclusively calm activities (such as eating out or playing mini-golf) reported back when they had arranged for a second date. In this group, 31 couples reported a second date planned an average of 3.3 days after the first, with a sample standard deviation of 1.2 days.
- Another random sample of couples on first dates with an adrenaline-filled component (such as go-karting, roller coasters, or karaoke). A total of 36 couples reported setting a second date an average of 2.1 days later, with a sample standard deviation of 0.9 days.
- Create a 90% confidence interval for the difference between the mean times to set a second date in the two groups (adrenaline daters and non-adrenaline daters).
- 13.** A study aims to examine whether or not individuals with chronic pain conditions (such as arthritis, irritable bowel syndrome, fibromyalgia,...) will generally have different systolic blood pressure than those without chronic pain. A random sample of 200 individuals suffering from chronic pain yielded a sample mean systolic pressure of 122 when examined during a period of relief. If we know that the true population standard deviation is 11, is there enough evidence here to suggest that those suffering from chronic pain will generally have a systolic pressure that is different from the norm of 120? Use a 2% level of significance.
- (a) (2 points) State the null and alternative hypotheses.
 - (b) (2 points) Calculate the test statistic.
 - (c) (2 points) Find (or estimate) the p -value for the test.

(d) (3 points) State and **interpret** your conclusion.

14. A study seeks to find a link, if any, between obsessive thoughts and supplementation with GABA amino acids. Participants (a random selection of individuals undergoing therapy with a record of obsessive thinking) complete a questionnaire that rates various aspects of their mental health on scales from 1 to 10 upon selection. The questions relating to obsessive thinking are examined and a score out of 80 is given to each participants, with higher scores being indications of more intrusive obsessive thinking. The patients then spend three months taking a daily GABA supplement, then complete the questionnaire a second time. The results are below:

Participant	1	2	3	4	5	6	7	8
Initial score (A)	67	74	61	59	57	60	71	54
New score (B)	52	72	53	60	51	63	69	55
Difference ($A - B$)	15	2	8	-1	6	-3	2	-1

Suppose that scores are normally distributed across all similar patients. Use a 3% significance level to test the hypothesis that scores related to obsessive thinking generally decrease with GABA supplementation.

- (a) (2 points) State the null and alternative hypotheses.
- (b) (2 points) Show that the sample standard deviation of the decrease in score is $s \approx 5.9281$. Provide detailed steps.
- (c) (2 points) Calculate the test statistic.
- (d) (2 points) Find (or estimate) the p -value for the test.
- (e) (3 points) State and **interpret** your conclusion.
15. (4 points) A store launches a campaign to collect feedback from its customers in the following way: An employee is staffed at the checkout line Monday to Friday, 8:00am-3:00pm, to invite all customers who make a purchase to provide feedback about their experience in-store. They are handed a card, which they are told they can fill out by hand if they wish and place in a sealed box near the exit, or they can scan the QR code printed on it to complete the form online. Unfortunately, a printing error causes the QR codes to redirect to an invalid web address, so the overall response rate they got was even lower than anticipated.
- Name two types of bias at play in this scenario. Briefly explain your answers.

ANSWERS

1. (a) 0.2833
(b) 0.4444
(c) 0.4667
(d) 0.1333
(e) No
2. (a) 0.3333
(b) 0.1
(c) 0.2278
(d) 0.2731
(e) 0.8049
3. 19 958 400
4. (a) 462
(b) 0.2727
5. (a) 0.36
(b) 0.875
6. (a) 0.052
(b) 0.722
(c) 1.838
7. (a) 0.1897
(b) 11.986 years
8. (a) 0.6179
(b) $n = 100 \geq 30$ OR X is normally distributed
(c) 0.9988
9. (a) 8.8
(b) 2.6199
(c) 0.3936
10. (a) 553
(b) [0.0079, 0.1254]
(c) C
11. [2.3475, 2.8525]
12. [0.7544, 1.6456]
13. (a) $H_0 : \mu = 120$; $H_a : \mu \neq 120$
(b) $z^* \approx 2.57$
(c) P -value = 0.0102
(d) Reject H_0 (There is sufficient evidence to claim that those with chronic pain have different systolic blood pressure.)
14. (a) $H_0 : \mu_d = 0$; $H_a : \mu_d > 0$
(b) Details of the work may vary, but it helps to start by finding that the mean difference is 3.5.
(c) $t^* \approx 1.670$
(d) $0.050 < P\text{-value} < 0.075$
(e) Fail to reject H_0 (There is insufficient evidence to claim that GABA supplementation generally decreases obsessive thinking.)
15. Multiple answers possible, including coverage bias, faulty measurement tool bias, and non-response bias.